How to use AI Code Assistant

"How to leverage AI Code Assistants with Agent.md, SpecKit, Skills"

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How is this doc organized

- Topic: Working effectively with AI Code Assistants with Agent, Specs, Skills
- white-labelled experiment = git@github.com:amitrajpurkar/sprboot_api_cktBrkr.git
- Sequence
  - Background on AI Agents, IDEs ... pages 3-5 : xxx minutes... 20%
  - About Speckit ... pages 6-9 : xxx minutes... 5%
  - About Skills... pages 10-11
  - About Agents.md ... pages 12-13
  - SDLC Roles with AI Agents ... pages 14-16
  - Ending thoughts ... pages 17-18 : xxx minutes... 5%
  - Journey of one experiment ... pages 19-33 : xxx minutes... 60%
  -

Background on AI Code Assistants

- AI does not replace a task; it augments and helps with automation; work is still done by the developer
- Developer is responsible for the code generated by the AI Assistant based on prompts provided
- Large number of Coding Assistants are available; all are fairly similar; its a developers choice to pick one as per their style and need
- These code assistants can provide in-line suggestions while the developer is writing code or they allow developers to chat with it "ask-mode", or use "plan-mode" to design an action plan before doing changes or use "code-mode" allowing AI to write code. (These modes are Windsurf specific; every Assistant provides similar modes)
- SpecKit is one tool that works with Code Assistants to generate the specifications using slash-commands; The developer uses a **structured interaction with AI** to reach the final outcome — No adhoc prompting
- This does not end at Specs.. **Coding along with AI Agents is changing Software development cycle itself; to retain + enrich software engineer practices, "detailing-requirements", "developing-clear-unambiguous specifications" is now at the heart of the process**

Estimate talk time: 3-5 minutes ?

Which IDE to use?

Large number of IDEs available with similar capabilities

- *All AI Code Assistants have similar capabilities*— it is developers choice which one to use
- Each one is similarly priced... around $20 per month
- At the moment Claude Code (Anthropic) is most feature reach; but you ideally need "Max-version" priced at $100-$200 per month to use it to full extent
- To name a few: Copilot, AWS CodeWhisperer, Tabnine, Cursor, Replit Ghostwriter, Codeium, Windsurf, Cody
- Each of these IDEs connect with different LLMs: Claude Sonnet, Claude Opus, GPT-5, Grok, o3, SWE to name a few— this depends on your token usage or your subscription

LLM models

Which ones to use? And when? — Its developers choice/ taste — try different models — below is one example

- How much control do you want?

- I want to control: Kind of tasks?
  - Simple / directed changes
    - Priority 1 — SWE-1.5; Claude Haiku 4.5
    - Priority 2: GPT 5.2 (low reasoning)
  - Larger/ Complex changes
    - Priority 1: Claude Sonnent 4.5
    - Priority 2: GPT 5.2 (medium)
    - Priority 3: Claude Opus 4.5

- AI should take control: Task Complexity?
  - Scoped/ routine tasks
    - Priority 1: Claude Sonnet 4.5
    - Priority 2: SWE-1.5, Gemini 3 Pro
    - Priority 3: Claude Opus 4.5
  - Very Complex
    - Priority 1: Claude Sonnet 4.5 (thinking)
    - Priority 2: GPT 5.2 (high), GPT 5.1 Codex-Max
    - Priority 3: Claude Opus 4.5

Background for SpecKit

Spec Kit is built around the idea of **Spec-Driven Development (SDD)** — a new AI-centric development workflow that **makes specifications the true source of truth** and uses them to *drive the generation of actual working code* rather than just guide it.

- 🧩 **Core Ideas**
- **"What before How"** – Emphasize *what* the system should do (the intent) before *how* it should do it.
- **Executable Specs** – Specifications aren't just documentation; they are refined and used to *generate implementation* through AI agents.
- **Structured Workflow with AI Agents** – There's a multi-stage workflow:
  - **Constitution** – Set project principles
  - **Specify** – Write the product spec
  - **Plan** – Add tech context (stack, architectural decisions)
  - **Tasks** – Break work into actionable steps
  - **Implement** – Have AI assistants build the code from tasks
- **AI-first orchestration** – Heavy reliance on AI for interpreting specs, planning, and executing tasks, with humans in a review/verification loop.
- 🧠 **Why This Matters**

Traditional dev is often *code-centric* — start coding and defer specifications/documentation. SDD flips that: *specs drive the entire creation process*, and code is synthesized to match the spec.

- SpecKit provides set of templates for AI Code Assistants (pull the latest).. use CLI to configure these templates for your Code Assistant or setup manually
- Philosophy: document the specification, plan, tasks very well before making code changes
- What this means for a developer: **"document** accurately before you code**"**

Dimension	Spec-Driven with AI	BDD with Human Developers
Primary Artifact	Rich structured specifications	Behavior scenarios (Gherkin tests)
Driver of Work	AI agents generating code	Humans writing and implementing tests
Feedback	AI-built code validated by humans	Tests failing/passing validate code
Alignment	Specification → implementation	Conversation → scenario → implementation
Automation Scope	High (from spec to code)	High for tests, lower for full code
Risk	Misinterpretation by AI	Miscommunication among humans

Estimate talk time: 3-5 minutes ?

Resources on Speckit

Handy links to documentation, explanation

- From the creators:
  - https://github.com/github/spec-kit
- Youtube videos:
  - https://www.youtube.com/watch?v=a9eR1xsfvHg. ; https://www.youtube.com/watch?v=SGHIQTsPzuY. ; https://www.youtube.com/watch?v=8jtIXRyGMQU
- Windsurf links:
  - https://docs.windsurf.com/windsurf/cascade/workflows
- Similar alternative
  - https://github.com/Fission-AI/OpenSpec

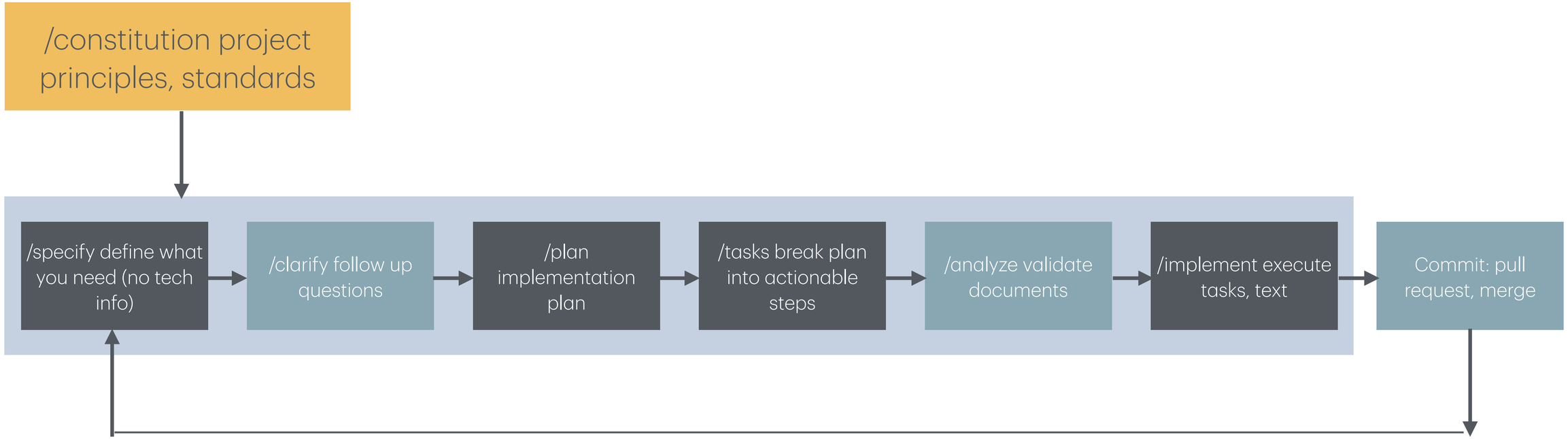
Comparing with OpenSpec

Two similar toolkits

Aspect	Speckit	OpenSpec
Purpose and Focus	Framework for writing and managing specifications	Open-source collaboration for specification-driven development
Integration with AI	Direct integrations with AI coding assistants	Similar integrations
Usability	Polished user interface for enhanced productivity	May require more technical knowledge; customizable
Community and Support	Typically backed by commercial support	Relies on community support; responsiveness varies
Flexibility and Extension	Structured extensibility for easy build-on	Encourages flexibility; open-source allows custom tweaks
Documentation, Learning Curve	Comprehensive documentation lowers learning curve	Depends on community contributions; may vary in quality

Command Workflow

- Basic: constitution —> specify —> plan —> tasks —> implement
- Alternate: constitution -> specify -> clarify -> plan -> tasks -> analyze -> implement



Things to know: Skills

When working with AI Coding Assistants

Dimension	GitHub spec-kit	Anthropic skills
Primary goal	Enable **spec-driven software development** with AI assistance	Extend an AI agent (Claude) with **modular, reusable capabilities**
Core philosophy	*"Specs are the source of truth"* → code follows specs	*"Teach the agent how to do things"* via skills
Level of abstraction	**Project / system lifecycle**	**Task / capability level**
View of AI	AI as a **co-developer** guided by structured specs	AI as an **agent** enhanced by learned skills
Intended outcome	More predictable, disciplined AI-assisted coding	Smarter, context-aware agent behavior

- spec-kit = **"what good looks like"**
- skills = **"how good work is done"**
- spec-kit is a process framework for AI-assisted software engineering
- skills is a capability framework for AI agents
- **spec-kit defines what should be built; skills define what the AI knows how to do.**

Dimension	spec-kit	skills
Drives *what*	The **entire software project**	A **specific task or behavior**
Drives *when*	Before and during development	During agent execution
Primary artifact	Specification documents	Skill definitions
Enforces workflow	Yes	No
Agent autonomy	Constrained by specs	Expanded by skills

Question	spec-kit answer	skills answer
"How do we build software with AI?"	Use specs to constrain and guide AI	Give AI better tools and instructions
"Where is control?"	Human-defined specs	Skill author's instructions
"Where is flexibility?"	Lower, by design	Higher, by design
"Risk mitigated?"	Scope creep, hallucinations, ambiguity	Lack of domain knowledge

SpecKit with Skills

These two can co-exist and complement each other

Layer	Tool	Responsibility
Product intent	spec-kit	Define goals, constraints, acceptance criteria
System design	spec-kit	Architecture, tradeoffs, non-functionals
Task decomposition	spec-kit	Break work into bounded tasks
Task execution	skills	Execute domain-specific actions
AI behavior	skills	Ensure consistent, repeatable agent behavior

Concern	spec-kit	skills
Scope control	✅	❌
Architectural consistency	✅	⚠️
Task quality	⚠️	✅
Behavioral consistency	❌	✅
Governance & review	✅	❌

Together → **controlled autonomy**.

- spec-kit governs what should be built and why
- skills govern how the AI performs specific actions
- Think of spec-kit as the contract and skills as the toolbox.

- Personally I have found Specs very useful, valuable; but I have been an architect for a long time, producing large amount of documentation myself

- When I interview developers , most complain that Specs are too “process-heavy” .. lots of documentation before they can write code

- This is where the tech-lead, principle developer should work with Business Analysts, Project manager, Quality Analyst to “write specs.md, run-clarify, write plan.md and tasks.md”

Agent.md

One more markdown you will hear

- **AGENT.md is an agent behavior contract.**
- It tells an AI coding assistant:

> *"When you operate inside this repository, this is who you are, how you behave, and what rules you must follow."*

It's **persistent context**, not a task, not a spec, not a capability.

Artifact	Controls	Applies to	Lifespan
AGENT.md	Behavior & discipline	AI agent	Long-lived
SpecKit	Intent & correctness	Features / systems	Per change
Skills	Capability & execution	Tasks	Reusable

```
┌──────────────────────┐
│ AGENT.md             │
│ (Behavior & rules)   │
│  • coding style      │
│  • boundaries        │
│  • safety            │
└──────────────────────┘
              │
              │ applies globally
              ▼
┌──────────────────────┐
│ SpecKit              │
│ (Intent & correctness)│
│  • requirements      │
│  • acceptance criteria│
│  • architecture      │
└──────────────────────┘
              │
              │ drives tasks
              ▼
┌──────────────────────┐
│ Skills               │
│ (Execution capability)│
│  • testing           │
│  • refactoring       │
│  • docs              │
└──────────────────────┘
```

- AGENT.md defines how the AI behaves
- Specs define what is being built
- Skills define what the AI can do

- AGENT.md is the constitution
- Specs are the laws
- Skills are the tools

- They're complementary — not redundant

AGENTS.md vs SPECS vs SKILLS

Which one fits where ??

Dimensions	Agents.md	Specs	Skills
	Define how the AI behaves	Define what is being built	Define what the AI can do
Primary purpose	Control AI Agents behavior consistently	Control the development process	Control specific type of task
Scope	Across projects, teams	Feature-based or for-a-system	For a task-type
Role	always-on	lifecycle-based	Task-based
Changes often?	rarely	Frequently, per feature/ project	Stays same for task-type
Audience	AI Agent	AI + Human	AI Agent
Enforces	Style, rules, discipline	Scope, intent, acceptance	How that type of task is done

So… the Developers part in this ??

“If AI is doing more of the typing, what must I still be good at so the system doesn’t collapse?”

- **Non-negotiable:** AI amplifies skills; it does not replace Developer’s skills
- **Architectural thinking:** decompose the system, decide which component, library or design pattern suites best for the problem at hand; think about transaction boundaries; interface contracts
- **Specify the problem:** translate business needs into technical intent; what should be the constraints; what should be the acceptance criteria
- **Review, Verification:** Because AI codes fast, you must validate faster; spot complexity; identify security holes; spot clever but wrong solutions; iterate through each step (its not fire-and-forget)
- Most imp — your **core development skills:** if you are a Java developer, you must know Java thoroughly; you should not work with half-baked-knowledge

For other roles in SDLC

Analysts, Testers, Project Managers, Architects...

The Need	Artifact
Behavioral governance	AGENT.md
Intent & correctness	Specs (SpecKit-style)
Execution capability	Skills

Those needs exist in **every SDLC role** — not just dev.

Role	Same pattern	Different surface
Dev	AGENT.md, Specs, Skills	IDE, repo
Analyst	Same	Jira, Confluence, docs
QA	Same	Test mgmt tools
UX	Same	Figma, design docs
PM	Same	Roadmaps, planning tools

The **pattern is universal**.
The **integration point differs**.

- Agent.md, Specs and Skills each generate good amount of documentation; though it appears to be dev-centric
- These three are not developer-only concepts; any role that works through structured artifacts and repeatable process can benefit
- Non-Developer roles can benefit more; their work is already text-heavy; many times artifacts are ambiguous
- A shared-intent, Spec between all roles is valuable; structured intent writing is important
- Standardize the intent, encode behavior, automate execution across all roles.

Planning to work with AI

- How I planned/ broke-down migration effort into phases and steps for making changes
- Sequencing and prioritization of the phases and steps
- Making the AI analyze the current codebase itself and match its analysis with my own experience creating the original app
- Getting the ground ready to let AI start making changes
- Key point —> we don’t start coding directly when using AI.. we need to plan, document and be very specific on what we need to do.

Ending thoughts

- AI is still evolving at a very rapid pace.. standardization in coding practice using AI is in its infancy
- Developers need to try the tools and experiment with different approaches as they surface.
- SpecKit is providing one such path; Claude Code is also using a very similar approach using “skills, scripts, templates, workflows” (comparing to memory, scripts, templates, workflows in SpecKit)
- If you don’t use concepts like Agent.md, Specs, Skills, the developer will need to babysit with AI for a long time and no consistency can be reached; **we need to bring repeatability, predictability into software engineering to make production code reliable**; Feeding the AI Agents with these three will help get us there

From other developers

Provide references to what other developers are researching

- Follow podcasts/ spotify/ YouTube from leading developer organizations: Thoughtworks, MartinFowler, Info-Q, ChangeLog, Software Engineering podcast; — plus the key sources: Anthropic, OpenAI, Google, Microsoft
- https://podcasts.apple.com/us/podcast/how-ai-will-change-software-engineering-with-martin-fowler/id1769051199?i=1000737446931
- https://martinfowler.com/articles/legacy-modernization-gen-ai.html
- https://www.thoughtworks.com/en-in/insights/podcasts/technology-podcasts/using-generative-ai-legacy-modernization
- https://youtu.be/6aD4c_izpnE?si=visMMXLY5mkJrQ8j
- https://youtu.be/yNH52oDpIKw?si=TFH8X0ErUW6E7-_O
- https://www.deeplearning.ai/. .... https://www.anthropic.com/learn
- Worth watching **—> https://youtu.be/wVyQn6QRcT4?feature=shared**

Experiment:

... migrate sample application from old to new stack

How can AI address Legacy App??

A **legacy application** is a software system that:

- Is **critical to business operations**, but
- Is **difficult, risky, or costly to modify, maintain, or replace**, due to technical, organizational, or strategic limitations.

> In practice, something becomes "legacy" when **change velocity slows down** while **business dependency remains high**.

- ❌ "Old software = legacy"
  → Not necessarily. Well-maintained older systems may not be legacy.
- ❌ "Legacy means useless"
  → Many legacy systems are **mission-critical** and extremely valuable.
- ❌ "We must rewrite everything"
  → Full rewrites are often the **riskiest option**.

Ask these questions:

- Is the system **still delivering business value**?
- How often does it need to change?
- What is the **risk of failure** vs. risk of change?
- Is the main pain **technology, architecture, people, or business fit**?
- What is the **time horizon** (1 year vs. 5 years)?

The answers usually determine the strategy.

- Qualities of Legacy application
  - Technical Characteristics: outdated tech, tight coupling, poor architecture/ design, low testability
  - Human factors: loss of institutional knowledge; skill scarcity
  - Operational Characteristics: high risk of change, costly maintenance
  - Business & Strategic factors: misaligned with current business needs, compliance/ security risks
- Different ways to address Legacy application
  - Leave as is
  - Refactor/ Modernize
  - Re-platform
  - Rewrite/ Replace
  - Strangler pattern
- Observation:
  - Investigate both code as well as business requirements; get latest business requirements; use AI to analyze -> document -> get specs; AI for wrap-around testing; tackle one feature at a time;

Estimate talk time: 3-5 minutes ?

About the Experiment

Migrating a sample application from old tech stack to new

- Why is this app legacy? a working application written in 2020 on older tech stack; newer versions of Java, Gradle do not support that stack; one of the key libraries is no longer supported today; a demo app with rest APIs which are expected to run 24/7 with a high transaction volume of up to 3,000 transactions per minute.

- This experiment was started before I knew about SpecKit; hence it used AI Assistant following a plan and set sequence of instructions

- Refactoring and modernization technique was used

- Key points:
  - Initial app was developed by myself in 2017; so the entire codebase as well as business requirements were well known; this was not a black-box
  - Target tech stack after upgrade was also known before hand
  - Experiment was with sequencing instructions to the AI Assistant and testing each step of the way

Glimpse of First Analysis

AI reviews, shows findings, Developer verifies

Pattern	Implementation	Quality
Circuit Breaker	Hystrix + Failsafe	⭐⭐⭐⭐⭐ Excellent dual implementation
Aspect-Oriented Programming	Spring AOP for logging	⭐⭐⭐⭐ Well-structured aspects
Builder Pattern	Product model	⭐⭐⭐⭐ Clean implementation
Configuration Properties	Type-safe config binding	⭐⭐⭐⭐⭐ Exemplary use of `@ConfigurationProperties`
Repository Pattern	Spring Data MongoDB	⭐⭐⭐⭐ Standard implementation
Dependency Injection	Constructor/field injection	⭐⭐⭐ Mixed approach (could be improved)

Package Organization:

```
com.anr/
├── Bootstrap.java          # Main application entry point
├── common/                 # Utilities (SBUtil)
├── config/                 # Configuration classes (9 files)
├── controller/             # REST controllers (4 files)
├── exception/              # Exception handling
├── localmdb/               # MongoDB domain & repositories
├── logging/                # AOP logging aspects (5 files)
├── model/                  # Response models
└── service/                # Business logic layer (3 files)
```

Technology	Version	Status	Assessment
Java	14.0.2	🟡 Outdated	LTS versions: 11, 17, 21 recommended
Spring Boot	2.3.3.RELEASE	🟡 Outdated	Current: 3.x (requires Java 17+)
Gradle	6.6	🟡 Outdated	Current: 8.x
Hystrix	2.2.4.RELEASE	🔴 Deprecated	Maintenance mode since 2018
Failsafe	1.1.0	🟡 Very old	Current: 3.x
MongoDB Driver	Spring Data	✅ Good	Properly abstracted
SpringDoc OpenAPI	1.4.4	🟡 Outdated	Current: 2.x
JaCoCo	0.8.5	🟡 Outdated	Current: 0.8.11+
ArchUnit	0.9.1	🔴 Very old	Current: 1.x
Lombok	Included	✅ Good	Reduces boilerplate
Gson	2.8.6	🟡 Outdated	Consider Jackson (already included)
Guava	29.0-jre	🟡 Outdated	Current: 32.x

- Ask AI Code Assistant to analyze the codebase
- As a developer, verify the findings documented by AI
- In my case, being the original developer of this application written in 2020, I was able to verify accuracy of the analysis

First Analysis continued

9.1 What's Good (Still Relevant in 2025)

Pattern/Practice	Status	Notes
Circuit Breaker Pattern	✅ Timeless	Core resilience pattern
AOP for Cross-Cutting Concerns	✅ Current	Still best practice
Type-Safe Configuration	✅ Current	Spring Boot standard
Repository Pattern	✅ Current	Clean architecture
Structured Logging	✅ Current	Observability requirement
API Documentation	✅ Current	OpenAPI standard

9.2 What's Outdated

Aspect	Issue	Modern Alternative
Hystrix	Deprecated	Resilience4j
Java 14	Non-LTS	Java 17 or 21 LTS
Spring Boot 2.3	Old	Spring Boot 3.x
JUnit 4	Legacy	JUnit 5 (Jupiter)
Mockito 1.x	Ancient	Mockito 5.x
WAR Packaging	Traditional	JAR + Container
Servlet Stack	Blocking I/O	Reactive (WebFlux) optional

- Even before I showed AI my plan for migration, it had listed correctly which features were good and which outdated.
- This also helped me update my own plan for migration

After migration was completed

Backend

Component	Technology	Version
Framework	Spring Boot	3.2.10
Language	Java	21
Build Tool	Gradle	9.1.0
Database	H2	2.2.224
ORM	JPA/Hibernate	-
API Docs	OpenAPI/Swagger	2.3.0
Circuit Breaker	Resilience4j	2.2.0

Frontend

Component	Technology	Version
Framework	React	18.2.0
HTTP Client	Axios	1.6.0
Build Tool	React Scripts	5.0.1
Styling	CSS3	-
State Management	React Hooks	-

Drafting the plan

- Step 1: ask AI to analyze codebase; describe what this application is and what key features it has (NOTE: this later part is because I was the original developer)
- Step 2: ask AI to create a step by step plan for migrating this application to latest stack and document this plan in a markdown file. Give it a hint to use gradle 9, java 21
- Step 3: ask AI to create separate migration plan for circuit breaker migration and save that in another markdown; give hint to use resilience4j (Here as well I had researched this lib before hand)
- Step 4: ask AI to create another plan for changing backend from mongodb to H2
- NOTE: separate research was done beforehand on steps 2, 3, 4; experiment was observing how AI was addressing these tasks

Journey of changes

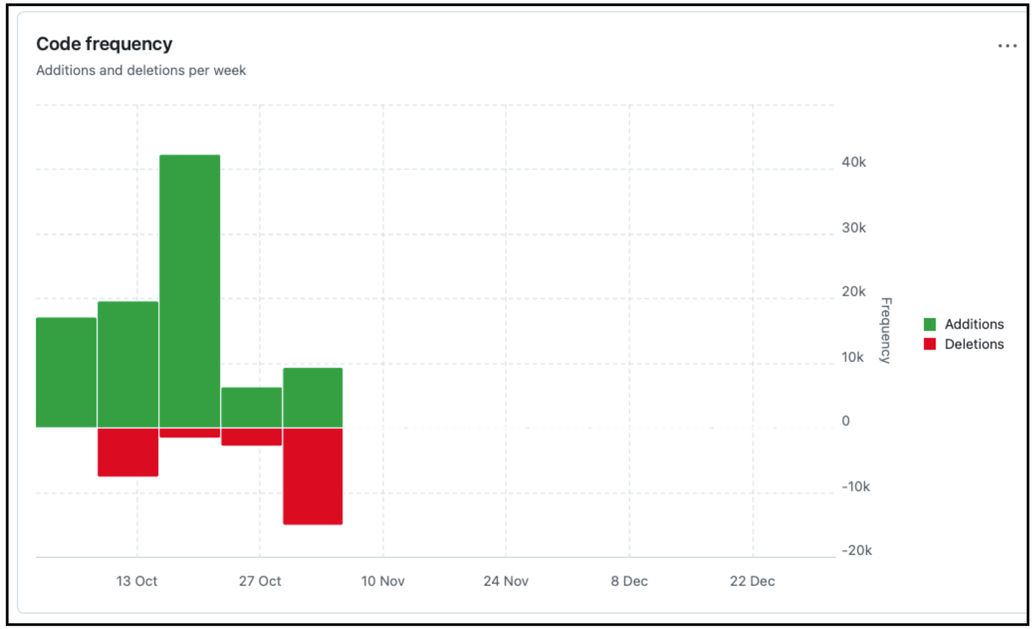
Comparing my approach with speckit

- Further experimenting on other greenfield projects using speckit; the sequential plan I used here was in line with speckits' specify, clarify, plan commands
- The features or changes needed were called out in steps 2, 3, 4
- Also fair amount of interaction went behind each steps to get it to a desired level: java/springboot-upgrade; circuit-breaker-upgrade; database-upgrade;
- One important distinction; in this experiment I was having AI implement each step fully at a time; unlike executing "task-command" followed by "implement";
- Also for each code change AI was making, I was verifying those changes, testing with unit tests and then committing changes that looked good.

Backend working with new stack

Compare with speckit

- In the migration experiment I had fair amount of interaction with AI to reach on a good enough upgrade; later experiments using speckit proved that if specification and plan is build with careful details, these iterations can be controlled
- It took me 3-4 weeks to complete steps 2, 3, 4 as outlined before. — speckit can help reduce this time
- Verifying changes made by AI using test cases was also important; on some occasions I also had AI generate more test cases. I was manually checking these in the Windsurf to see how these tests ran.



Increase testability

- My initial test cases in the original application were only checking on the service layer, repository layer, and Aspects
- With code changes made by AI I started adding more tests around utilities, service layer and even controller
- Next I used AI to increase my test coverage to a higher level than before
- Classifying test cases per feature using BDD style test cases is helpful for pair programming with AI; more so as you are detailing fine-grain specifications; so its better to test is the code is satisfying those specifications.

Adding new feature

- This app was not having a client for the API — either a web-frontend or a CLI; so next came feature additions now that the essential upgrade was done.
- I asked AI to add a frontend UI in ReactJS and that it should be a single page application
- This also ran through numerous iterations till desirable frontend UI was working

Adding documentation for the app

- Over the 5 week period working with AI to migrate the legacy application to new tech stack, I had been making my notes each day. After both backend and frontend were working as desired, now was a good time to capture all notes in the repository itself

- I had AI generate documentation, of my journey — project summary, architecture, migration guides for gradle, resilience4j, H2; implementation guides, test summaries, code review, quick start guides, etc

Re-arrange docs to Mkdocs

- For my experiment, I hand crafted the mkdocs for this application using documentation generated by AI
- I arranged the pages, the navigation, wrote my own index page and few others to organize documentation in certain way
- With thoughtful planning this could be achieved by asking AI generate documents in this layout and structure.

Summarizing this experiment

- As my first experiment playing with AI Code Assistant, this did give me insight on how I should be interacting with AI
- I experimented with SpecKit separately on simple example projects from scratch
- If I need to experiment on a large legacy system and also make use of SpecKit; my personal approach will be to isolate one feature or aspect of that legacy app and work with AI using SpecKit to understand and document the behavior of that feature; verify with business partners before making any changes.

Appendix

... extra pages

Old advice is still relevant

“Working effectively with Legacy Code” by Michael Feathers (2004)

- First add tests and then do your changes
- Identify places to alter program behavior without changing code (“seams”)
- Create tests that characterizes behavior of the piece of code
- Use sprout and wrap techniques
- Sprout: create code in separate module/ package; test it; identify insertion point; call it from legacy code
- Wrap: rename old method you want to wrap; create a new method with same signature as old; call old method from new; put logic before & after call; like AOP